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Cross Platform App Dev Lab 2

AIM - Develop different layout structures using Row, Column, and Stack widgets in Flutter

THEORY -

Flutter provides several layout widgets that are used to arrange UI elements efficiently. Among the most commonly used layout widgets are **Row**, **Column**, and **Stack**. These widgets help developers create responsive and well-structured user interfaces.

1. Row Widget

The **Row** widget arranges its child widgets **horizontally**, from left to right by default. It is useful when multiple elements need to be displayed in a single horizontal line, such as buttons, icons, menus, or images.

Important properties:

- `mainAxisAlignment` – controls horizontal alignment of children.
- `crossAxisAlignment` – controls vertical alignment of children.
- `mainAxisSize` – determines how much horizontal space the Row occupies.
- `children` – contains the widgets to be displayed.

2. Column Widget

The **Column** widget arranges its child widgets **vertically**, from top to bottom by default. It is commonly used for forms, lists of information, login screens, and vertically organized UI components.

Important properties:

- `mainAxisAlignment` – controls vertical alignment of children.
- `crossAxisAlignment` – controls horizontal alignment of children.
- `mainAxisSize` – determines how much vertical space the Column occupies.
- `children` – contains the widgets to be displayed.

3. Stack Widget

The **Stack** widget places its children **on top of one another**, allowing widgets to overlap.

It is useful for creating layouts such as images with text overlays, badges on icons, profile pictures with status indicators, and floating elements.

The **Positioned** widget can be used inside a Stack to specify the exact position of a child.

4. Difference Between Row, Column, and Stack

Widget	Arrangement	Common Use
Row	Horizontal	Menus, buttons, icons
Column	Vertical	Forms, text sections, screens
Stack	Overlapping	Image overlays, badges, floating elements

5. Axis Concept

Row and Column use two axes -

- **Main axis** – the direction in which children are arranged.
- **Cross axis** – the direction perpendicular to the main axis.

For a **Row**, the main axis is horizontal and the cross axis is vertical.

For a **Column**, the main axis is vertical and the cross axis is horizontal.

APP OVERVIEW

App title - Campus Companion

A student-focused campus information dashboard built as a Flutter MVP for LY B.Tech Computer Engineering (VJTI). It demonstrates clean UI design and meaningful usage of Flutter's core layout widgets: Row, Column, and Stack.

Key Features -

1. Home Dashboard:

Displays a personalized greeting, date, quick snapshot of today's classes, and a 6-feature grid (3 active + 3 "Coming Soon" badges).

2. Weekly Timetable:

Features a day selector (MON–FRI) populated with the actual AY 2026–27 LY CE timetable, displaying class timings, subjects, faculty, and room/lab numbers.

3. Professor Directory:

Lists CE department faculty with a detail view for room locations, office hours, email, and subjects taught.

4. Campus Events:

Displays upcoming campus events (Tech, Cultural, Hackathons) with detailed views and category overlays.

Usage of Row, Column, and Stack Widgets

1. Column (Vertical Alignment)

- Column is used to structure overall screen layouts, forms, list items, and stacked text blocks.
- **Screen Layouts:**

- HomeScreen: Vertically stacks the header greeting, Today's Classes card, section title, and feature grid.
- TimetableScreen: Vertically arranges the department banner, day selector, and class list.
- ProfessorDetailScreen: Stacks the profile card, contact section, and subject list vertically.
- **Card & Text Hierarchies:**
 - FeatureCard: Arranges icon, title, and description vertically.
 - ClassCard: Arranges subject title and professor info in a vertical block.

2. Row (Horizontal Alignment)

- Row is used for headers, side-by-side metadata items, chip selectors, and horizontal card splits.
- **Headers & Controls:**
 - HomeScreen: Greeting text and profile avatar side-by-side.
 - DaySelector: Horizontally places MON | TUE | WED | THU | FRI selection chips.
- **Data Fields & Card Layouts:**
 - ClassCard: Places start/end time column, vertical divider line, subject details, and room badge in a horizontal sequence.
 - ProfessorCard: Aligns faculty avatar icon, name/department info, and navigation chevron arrow across a row.
 - EventCard: Displays date, time, and location icons with their text side-by-side using inner rows.

3. Stack (Overlapping & Badge Positioning)

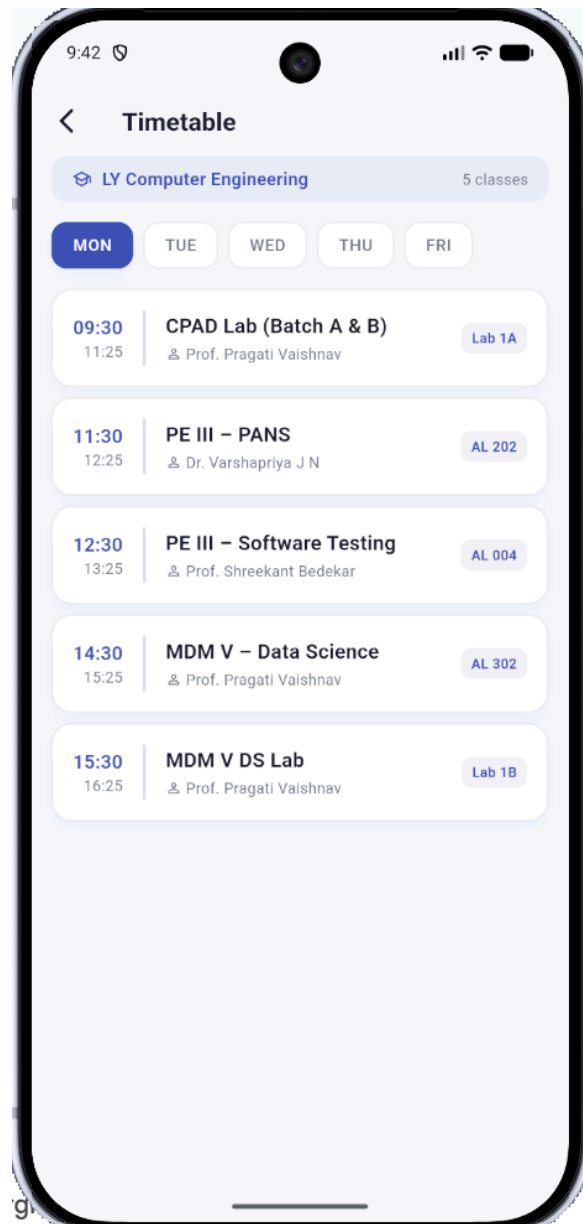
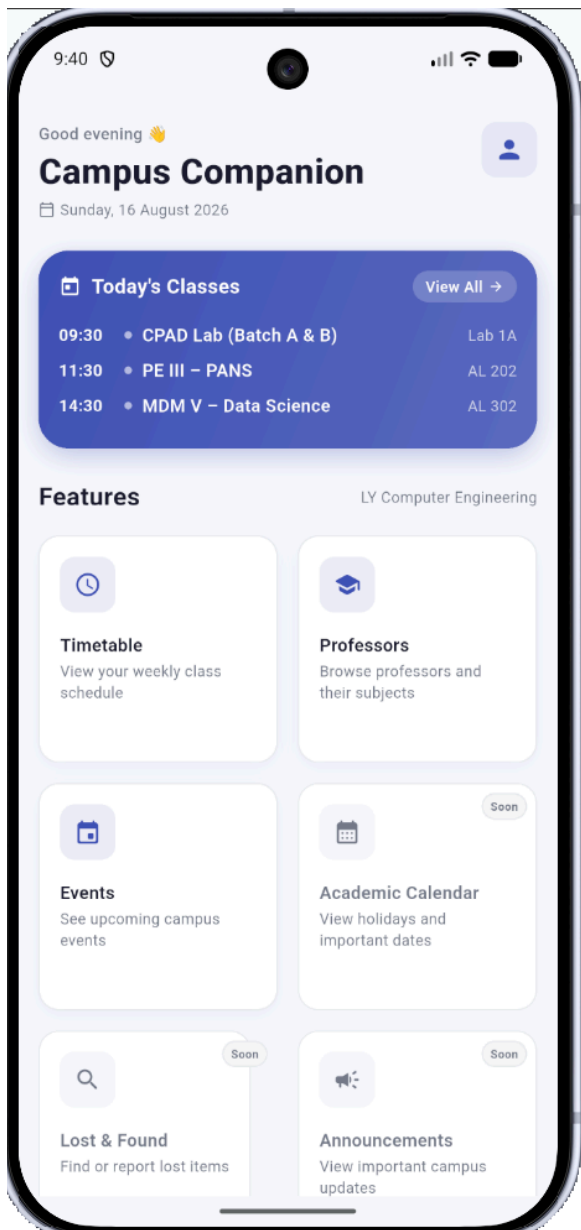
- Stack is used specifically where elements need to visually float or overlap on top of base containers.
- **Coming Soon Badges (FeatureCard):**
 - FeatureCard: Overlays a "Soon" pill badge at the top-right corner (Positioned) of disabled cards without disrupting the inner card content padding.
- **Category Badges (EventCard):**
 - EventCard: Overlays color-coded event category badges (Tech, Cultural, Hackathon) at the top-right position of event cards.

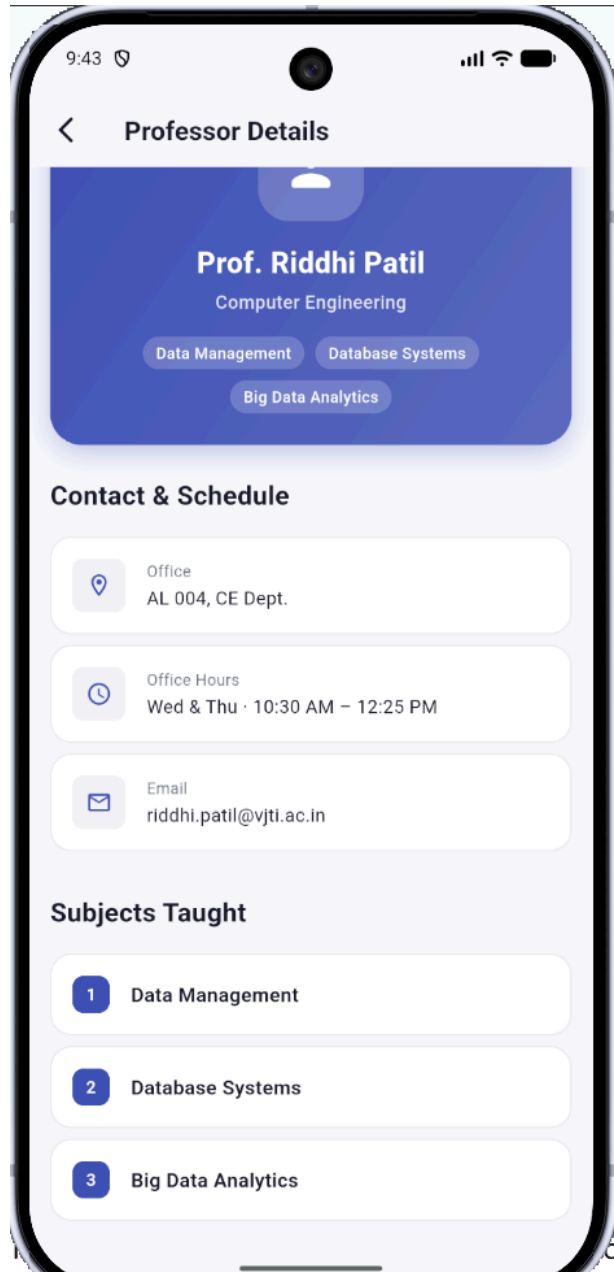
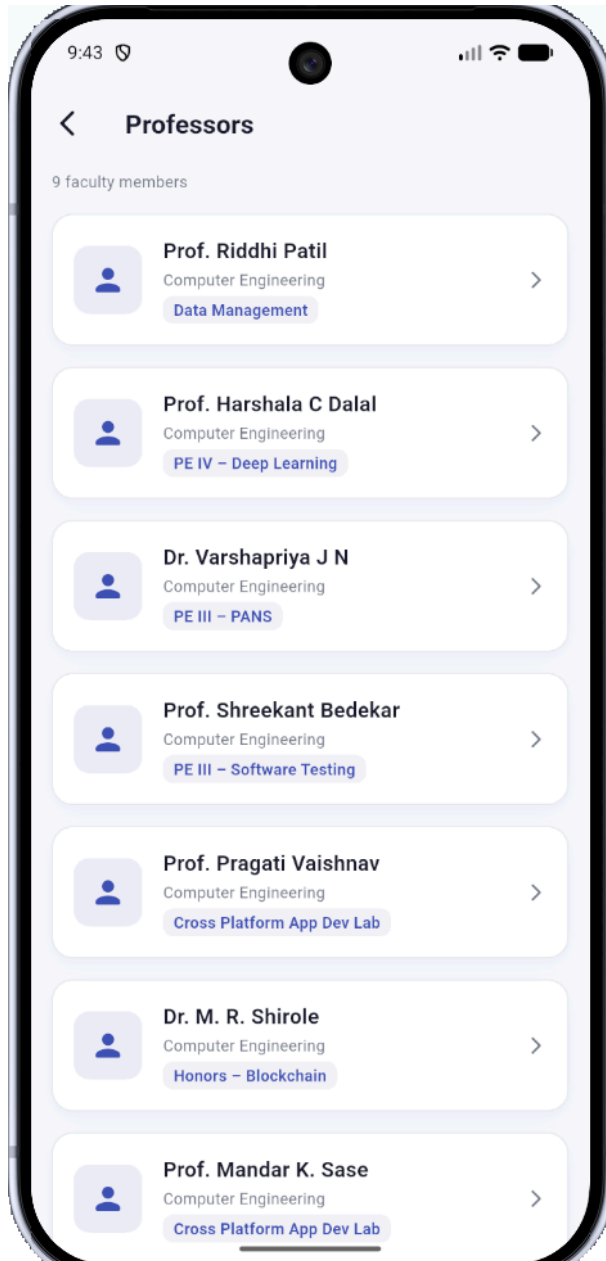
- **Hero Event Banner (EventDetailScreen):**
 - EventDetailScreen: Overlays a translucent category tag over the top-right of the hero gradient header.

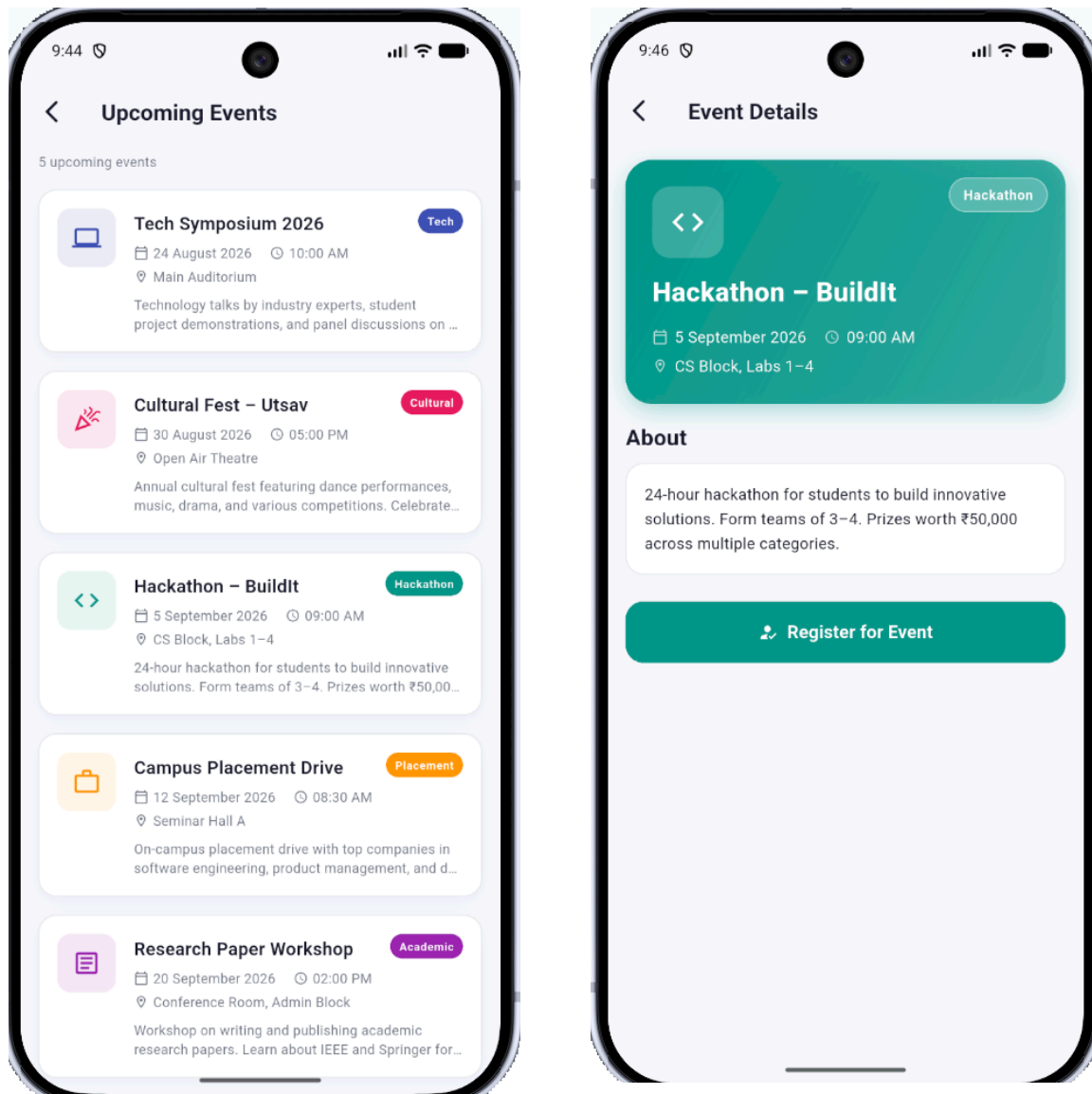
REPO LINK FOR CODE -

https://github.com/YashviMehta03/CPAD_lab_repos/tree/main/lab2_app

SCREENSHOTS -







CONCLUSION -

This lab provided a practical understanding of core Flutter layout widgets, specifically Row, Column, and Stack. By implementing the "Campus Companion" app, I gained hands-on experience in arranging UI elements horizontally, vertically, and through overlapping layers, which is essential for building responsive and structured interfaces. Mastering these fundamental widgets has strengthened my ability to translate complex UI designs into functional, clean, and organized Flutter applications.